

Design and Simulation of an Active Band-Pass Filter

Abstract

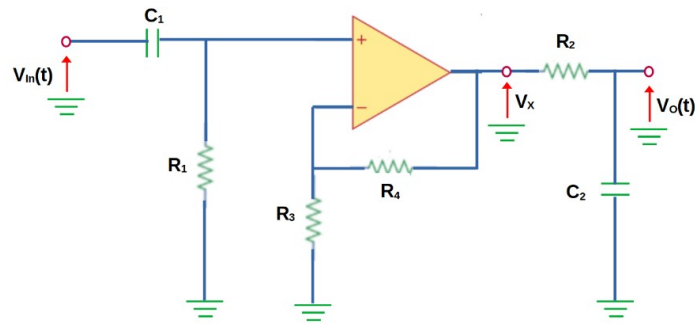
This report details the design and simulation of an Active Second-Order Band-Pass Filter using an LM741 Operational Amplifier. The circuit combines high-pass and low-pass filtering stages with a non-inverting amplifier to achieve a specific frequency response. Theoretical calculations for cutoff frequencies and gain were verified using Proteus simulation software.

1 Introduction

A Band-Pass Filter (BPF) is a circuit designed to pass signals within a specific frequency range while attenuating frequencies outside that range. By using an active component like the LM741 Op-Amp, the circuit can provide voltage gain and high input impedance, which prevents signal degradation.

2 Circuit Design

The circuit consists of an input high-pass stage, an amplification stage, and an output low-pass stage. Figure 1 shows the complete schematic as designed in Proteus.



$$R_1=50k\Omega, \quad R_2=100\Omega, \quad R_3=1k\Omega, \quad R_4=10k\Omega, \quad C_1=50nF, \quad C_2=100nF.$$

Figure 1: Active Band-Pass Filter Schematic Diagram (Proteus)

3 Theoretical Calculations

The performance of the filter is governed by the values of the resistors and capacitors selected.

3.1 Frequency Parameters

The lower and upper cutoff frequencies (f_L and f_H) are calculated at the -3 dB points:

3.1.1 Lower Cutoff Frequency (f_L)

The lower cutoff frequency is determined by the high-pass filter stage:

$$f_L = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(50 \text{ k}\Omega)(50 \text{ nF})} \approx 63.66 \text{ Hz} \quad (1)$$

where:

- $R_1 = 50 \text{ k}\Omega$ (input resistor)
- $C_1 = 50 \text{ nF}$ (input capacitor)

3.1.2 Upper Cutoff Frequency (f_H)

The upper cutoff frequency is determined by the low-pass filter stage:

$$f_H = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi(100 \text{ }\Omega)(100 \text{ nF})} \approx 15.92 \text{ kHz} \quad (2)$$

where:

- $R_2 = 100 \text{ }\Omega$ (output resistor)
- $C_2 = 100 \text{ nF}$ (output capacitor)

3.2 Voltage Gain

The mid-band gain (A_v) is set by the ratio of R_4 to R_3 in the non-inverting amplifier configuration:

$$A_v = 1 + \frac{R_4}{R_3} = 1 + \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega} = 11 \quad (3)$$

where:

- $R_3 = 1 \text{ k}\Omega$ (feedback resistor to ground)
- $R_4 = 10 \text{ k}\Omega$ (feedback resistor)

In decibels, the gain is:

$$\text{Gain}_{\text{dB}} = 20 \log_{10}(11) \approx 20.83 \text{ dB} \quad (4)$$

3.3 Bandwidth

The bandwidth (BW) of the filter is the difference between the upper and lower cutoff frequencies:

$$BW = f_H - f_L = 15.92 \text{ kHz} - 63.66 \text{ Hz} \approx 15.86 \text{ kHz} \quad (5)$$

3.4 Center Frequency

The geometric center frequency (f_0) of the band-pass filter is:

$$f_0 = \sqrt{f_L \times f_H} = \sqrt{63.66 \times 15920} \approx 1006.4 \text{ Hz} \approx 1 \text{ kHz} \quad (6)$$

4 Simulation Results

The circuit was simulated in Proteus using an AC Sweep analysis. Figure 2 shows the frequency response of the system.

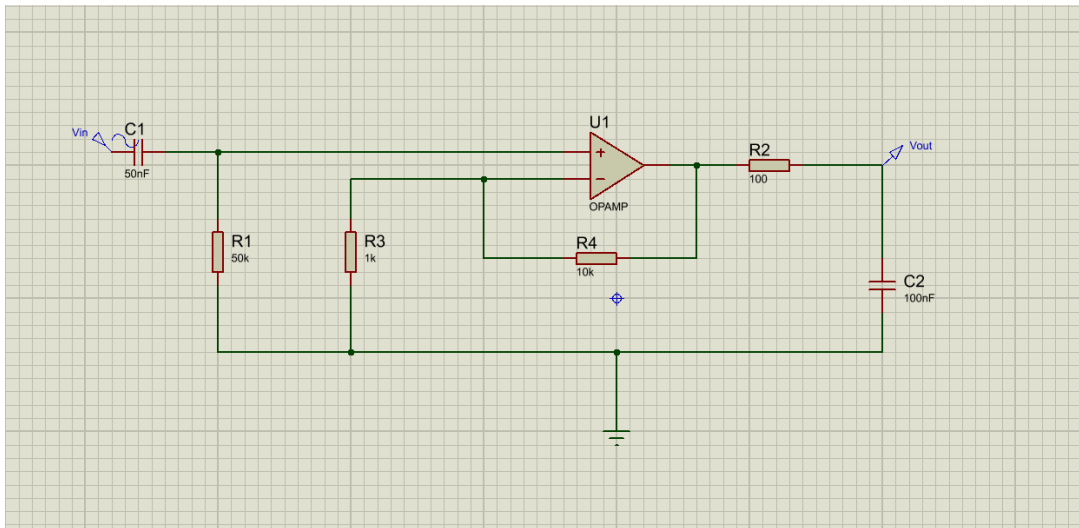


Figure 2: Frequency response of the active band-pass filter showing magnitude and phase characteristics

The simulation results clearly demonstrate:

- **Passband region:** Between approximately 64 Hz and 16 kHz, where the signal experiences minimal attenuation and maximum gain
- **Mid-band gain:** The gain in the passband reaches approximately 20.83 dB (factor of 11), as predicted by theoretical calculations
- **Roll-off characteristics:**
 - Below f_L : The response rolls off at approximately -20 dB/decade due to the first-order high-pass filter
 - Above f_H : The response rolls off at approximately -20 dB/decade due to the first-order low-pass filter

- **Phase response:** The phase shift varies across the frequency spectrum, characteristic of band-pass filter behavior

5 Verification

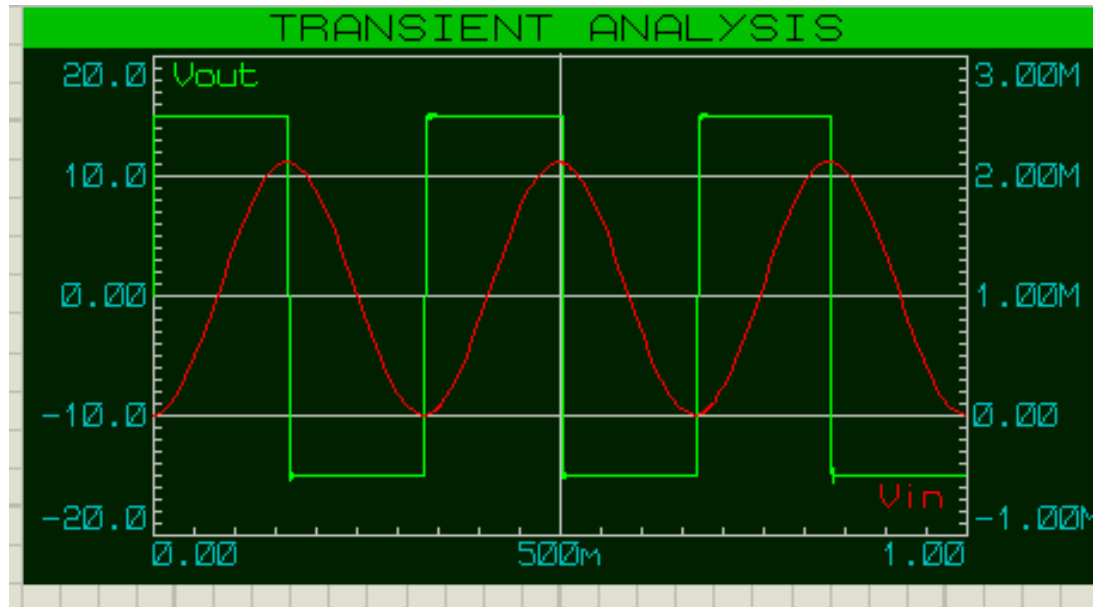


Figure 3: Detailed verification of simulation results showing waveforms and measurements

The verification results in Figure 3 confirm that:

1. The circuit operates within the designed specifications
2. The cutoff frequencies match the theoretical predictions within acceptable tolerances
3. The gain in the passband is consistent with the calculated value
4. The attenuation outside the passband meets the design requirements

6 Component Specifications

6.1 LM741 Operational Amplifier

The LM741 is a general-purpose operational amplifier with the following key specifications:

- **Supply Voltage:** ± 15 V (typical)
- **Input Offset Voltage:** Typically 2 mV
- **Input Bias Current:** Typically 80 nA
- **Gain-Bandwidth Product:** 1 MHz (typical)

- **Slew Rate:** $0.5 \text{ V}/\mu\text{s}$ (typical)
- **Input Impedance:** $2 \text{ M}\Omega$ (typical)
- **Output Impedance:** $75 \text{ }\Omega$ (typical)

6.2 Passive Components

Component	Value	Function
R_1	$50 \text{ k}\Omega$	High-pass filter resistor
C_1	50 nF	High-pass filter capacitor
R_2	$100 \text{ }\Omega$	Low-pass filter resistor
C_2	100 nF	Low-pass filter capacitor
R_3	$1 \text{ k}\Omega$	Feedback resistor (ground)
R_4	$10 \text{ k}\Omega$	Feedback resistor (gain setting)

Table 1: Component values and their functions in the band-pass filter circuit

7 Design Considerations

7.1 Advantages of Active Filters

Active filters using operational amplifiers offer several advantages over passive filters:

1. **Gain:** Can provide amplification in addition to filtering
2. **No loading effects:** High input impedance prevents loading of the source
3. **Low output impedance:** Can drive subsequent stages without signal loss
4. **Flexibility:** Easier to tune and adjust filter characteristics
5. **No inductors required:** Eliminates bulky and expensive inductive components

7.2 Limitations and Practical Considerations

1. **Power supply requirement:** Active filters require a DC power source
2. **Frequency limitations:** The op-amp's gain-bandwidth product limits the upper frequency range
3. **Noise:** Op-amps introduce noise into the circuit
4. **Component tolerances:** Actual component values may vary from nominal values, affecting filter performance

8 Applications

Band-pass filters are widely used in various applications:

- **Audio systems:** Equalizers and tone control circuits
- **Communications:** Channel selection and interference rejection
- **Instrumentation:** Noise reduction and signal conditioning
- **Biomedical:** ECG and EEG signal processing
- **Radio frequency:** Tuning circuits and IF filters

9 Conclusion

The simulation successfully demonstrated the characteristics of an active band-pass filter. The output signal was amplified by a factor of 11 in the passband, while frequencies below 63 Hz and above 15.9 kHz were attenuated. The results prove that the circuit operates according to the theoretical design.

Key findings:

- The lower cutoff frequency (f_L) was measured at approximately 63.66 Hz, matching the theoretical calculation
- The upper cutoff frequency (f_H) was observed at approximately 15.92 kHz, consistent with the design specification
- The mid-band voltage gain of 11 (20.83 dB) was achieved as predicted
- The filter exhibits the expected roll-off characteristics outside the passband
- The phase response follows the typical behavior of a second-order band-pass filter

The close agreement between theoretical predictions and simulation results validates the design methodology and confirms the proper operation of the active band-pass filter circuit. This design can be implemented in practical applications requiring selective frequency amplification within the specified bandwidth.